

US TOY SALES PERFORMANCE ANALYSIS PROJECT



PRESENTED BY : KUSHANG BHATTACHARYA

PROJECT WORKFLOW

DATA CLEANING & PRE-PROCESSING (MYSQL)

**EXPLORATORY DATA ANALYSIS & BUSINESS
QUERIES (MYSQL)**

**DATA MODELING & CUSTOM TIME INTELLIGENCE
(POWER BI)**

**INTERACTIVE DASHBOARDING & INSIGHT
DELIVERY (POWER BI)**

TECH STACK USED

- **MYSQL**
- **POWER BI**
- **DAX**



DATA CLEANING & PRE-PROCESSING (MYSQL)

- **REMOVED NULLS & BLANKS:** CHECKED ALL TABLES (SALES, PRODUCTS, STORES, INVENTORY) AND FIXED MISSING DATA TO KEEP CALCULATIONS ACCURATE.
- **REMOVED DUPLICATES:** CLEANED OUT REPEATING ROWS AND DOUBLE ENTRIES FROM IDS TO AVOID COUNTING THE SAME SALE TWICE.
- **FIXED DATA TYPES:** CONVERTED TEXT DATES INTO PROPER SQL DATE FORMAT AND CHANGED PRICE/COST COLUMNS INTO NUMBERS FOR CORRECT ROUNDING.

SQL QUERIES-DATA CLEANING

```
1 • CREATE DATABASE toy_store;
2 • USE toy_store;
3
4 • CREATE TABLE sales (
5     Sales_ID INT,
6     Date VARCHAR(50),
7     Store_ID INT,
8     Product_ID INT,
9     Units INT
10 );
11
12 • LOAD DATA LOCAL INFILE "C:/Users/rambh/Downloads/US_Toy_datasets/sales.csv"
13 INTO TABLE sales
14 FIELDS TERMINATED BY ','
15 ENCLOSED BY '"'
16 LINES TERMINATED BY '\n'
17 IGNORE 1 LINES;
18
19 # DATA CLEANING
20
21 # CLEAN THE SALES TABLE
22
23 • SELECT Date FROM sales LIMIT 5;
24
25 • SET SQL_SAFE_UPDATES =0;
26
27 • ALTER TABLE sales
28     MODIFY COLUMN Date DATE;
```

```
29
30 • SELECT * FROM sales
31     WHERE Sales_ID IS NULL
32     OR Date IS NULL
33     OR Store_ID IS NULL
34     OR Product_ID IS NULL
35     OR Units IS NULL;
36
37 • SELECT Sales_ID ,COUNT(*) FROM sales
38     GROUP BY Sales_ID
39     HAVING COUNT(*) >1;
40
41 # CLEAN THE PRODUCTS TABLE
42
43 • SELECT * FROM products
44     WHERE Product_ID IS NULL
45     OR Product_Name IS NULL
46     OR Product_Category IS NULL
47     OR Product_Cost IS NULL
48     OR Product_Price IS NULL;
49
50 # CLEAN THE STORES TABLE
51
52 • ALTER TABLE stores
53     MODIFY COLUMN Store_Open_Date DATE;
54
55 • UPDATE stores
56     SET Store_Name = REPLACE(Store_Name, ' 1','');
57
58 • SET SQL_SAFE_UPDATES =1;
```

SQL BUSINESS QUERIES & INSIGHTS

1. WHICH PRODUCT CATEGORY MAKES THE HIGHEST TOTAL REVENUE, AND WHICH CATEGORY HAS THE HIGHEST PROFIT MARGIN?

OUTPUT:

- **MEGA BOARD GAME (ACTION FIGURES CATEGORY) IS THE TOP PERFORMER, GENERATING A MASSIVE REVENUE OF \$231,910.47 AND A NET PROFIT OF \$110,006.94.**
- **CITY BOARD GAME (STEM KITS) FOLLOWS AS THE SECOND-HIGHEST REVENUE GENERATOR WITH \$107,791.60.**

```
3  # TOTAL REVENUE AND PROFIT BY PRODUCT NAME AND PRODUCT CATEGORY
4
5  •  SELECT
6  products.Product_Name,
7  products.Product_Category,
8  ROUND(SUM(sales.Units*products.Product_Price),2) AS Total_Revenue,
9  ROUND(SUM(sales.Units*(products.Product_Price - products.Product_Cost)),2) AS Total_Profit
10 FROM sales
11 INNER JOIN products
12 ON sales.Product_ID = products.Product_ID
13 GROUP BY products.Product_Name , products.Product_Category
14 ORDER BY Total_Revenue DESC;
```

Product_Name	Product_Category	Total_Revenue	Total_Profit
Mega Board Game	Action Figures	231910.47	110006.94
City Board Game	STEM Kits	107791.6	51186.96
Pro Science Lab	Outdoor Play	106205.54	43971.54
Neo Science Lab	Pretend Play	99060.64	36901.92
City Action Hero	Outdoor Play	97635.65	50024.45
Rainbow Plush Bear #70	Pretend Play	96809.44	35804.32
Deluxe Card Deck #161	Card Games	94478.02	44485.47
Neo Board Game	Puzzles	93533.44	49125.44
Rainbow Drone Toy	Action Figures	92972.08	38704.46
Mini Train Set	Card Games	92022	39126.5
Mega Art Kit #56	Card Games	89092.64	46905.3
Ocean Stacking Blocks	Board Games	88660.45	38790.51
Galaxy Music Toy	STEM Kits	88527.88	39172.16
City Plush Bear #42	Puzzles	87624.57	37594.7

SQL BUSINESS QUERIES & INSIGHTS

2. HOW MUCH TOTAL REVENUE DID EACH STORE MAKE BASED ON ITS CITY, AND WHAT IS THE AVERAGE NUMBER OF UNITS SOLD PER TRANSACTION?

OUTPUT:

- **SUNSHINE TOY STORE IN ARLINGTON, TX LEADS NATIONWIDE WITH A TOTAL REVENUE OF \$103,824.10.**
- **THE AVERAGE UNITS SOLD PER TRANSACTION IS HIGHLY UNIFORM AT 2.3 TO 2.4 UNITS ACROSS ALL TOP 10 STORES, INDICATING CONSISTENT CUSTOMER BUYING VOLUME.**

```
16 # TOP 10 STORES BY REVENUE AND AVG UNIT SOLD
17 • SELECT stores.Store_Name, stores.Store_City,
18    ROUND(SUM(sales.Units*products.Product_Price),2) AS Total_Revenue ,
19    ROUND(AVG(sales.Units),1) AS Average_Unit_Sold_per_Transaction
20 FROM sales
21 INNER JOIN stores
22 ON sales.Store_ID = stores.Store_ID
23 INNER JOIN products
24 ON sales.Product_ID = products.Product_ID
25 GROUP BY stores.Store_Name , stores.Store_City
26 ORDER BY Total_Revenue DESC
27 LIMIT 10;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:	Fetch rows:
Store_Name	Store_City	Total_Revenue	Average_Unit_Sold_per_Transaction	
▶ Sunshine Toy Store Arlington	Arlington, TX	103824.1	2.3	
Bluebird Toys Lexington	Lexington, KY	102623.94	2.4	
Toy Haven Akron	Akron, OH	102152.67	2.3	
Maple Street Toys Louisville	Louisville, KY	102045.91	2.3	
Mountain Trail Toys Pittsburgh	Pittsburgh, PA	101236.05	2.3	
Playtime Emporium Fort Worth	Fort Worth, TX	101151.19	2.3	
Maple Street Toys Des Moines	Des Moines, IA	99965.25	2.3	
Creative Play Toys Huntsville	Huntsville, AL	98942.13	2.3	
Cedar Toy Corner Montgomery	Montgomery, AL	98668.57	2.3	
Neighborhood Toy Co. Worcester	Worcester, MA	98042.97	2.3	

SQL BUSINESS QUERIES & INSIGHTS

3.WHAT IS THE MONTH-ON-MONTH SALES TREND? IN WHICH MONTH DO SALES PEAK, AND IN WHICH MONTH ARE THEY THE LOWEST?

OUTPUT:

- **PEAK PERFORMANCE: MARCH ACHIEVED THE HIGHEST SALES WITH A TOTAL REVENUE OF \$843,263.00.**
- **LOWEST PERFORMANCE: FEBRUARY RECORDED THE LOWEST SALES DIPPING TO \$763,977.05, SHOWING A SHARP RECOVERY TREND AS SPRING BEGINS.**

```
31 # TOP MONTH - MARCH
32 • SELECT
33 MONTHNAME(sales.Date) AS Month_Name,
34 ROUND(SUM(sales.Units*products.Product_Price),2) AS Total_Revenue
35 FROM sales
36 INNER JOIN products
37 ON sales.Product_ID = products.Product_ID
38 GROUP BY Month_Name
39 ORDER BY Total_Revenue DESC
40 LIMIT 1;
41
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows

	Month_Name	Total_Revenue
▶	March	843263

```
42 # LOWEST MONTH - FEBRUARY
43 • SELECT
44 MONTHNAME(sales.Date) AS Month_Name,
45 ROUND(SUM(sales.Units*products.Product_Price),2) AS Total_Revenue
46 FROM sales
47 INNER JOIN products
48 ON sales.Product_ID = products.Product_ID
49 GROUP BY Month_Name
50 ORDER BY Total_Revenue
51 LIMIT 1;
52
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows

	Month_Name	Total_Revenue
▶	February	763977.05

SQL BUSINESS QUERIES & INSIGHTS

4. WHICH HIGH-DEMAND PRODUCTS HAVE CRITICALLY LOW INVENTORY (STOCK ON HAND LESS THAN 10 UNITS) AND NEED URGENT REORDERING?

OUTPUT:

- **HIGH-DEMAND PRODUCTS LIKE MINI ROBOT (24,093 UNITS SOLD), GALAXY TRAIN SET (21,005 UNITS SOLD), AND RETRO ART KIT (20,709 UNITS SOLD) HAVE A REMAINING STOCK OF 0. THIS CRITICAL STOCKOUT MEANS THE COMPANY IS ACTIVELY LOSING OUT ON IMMEDIATE CUSTOMER DEMAND AND REVENUE.**

```
53 # INVENTORY ANALYSIS
54 • SELECT
55 products.Product_Name,
56 inventory.Stock_On_Hand,
57 SUM(sales.Units) AS Total_Units_Sold
58 FROM sales
59 INNER JOIN inventory
60 ON inventory.Product_ID = sales.Product_ID
61 INNER JOIN products
62 ON products.Product_ID = sales.Product_ID
63 WHERE inventory.Stock_On_Hand < 10
64 GROUP BY products.Product_Name,inventory.Stock_On_Hand
65 ORDER BY Total_Units_Sold DESC;
66
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content:

	Product_Name	Stock_On_Hand	Total_Units_Sold
▶	Mini Robot	0	24093
	Galaxy Train Set	0	21005
	Retro Art Kit	0	20709
	Pro Board Game	0	19491
	Glow Explorer Kit	0	19200
	City Action Hero	0	19110
	Adventure Train Set	0	18545
	Mega Explorer Kit	0	18270
	Neo Princess Doll #49	0	17960
	Neo Art Kit	0	16254

SQL BUSINESS QUERIES & INSIGHTS

5. DO CUSTOMERS BUY EXPENSIVE PRODUCTS IN SINGLE QUANTITIES AND CHEAP PRODUCTS IN BULK? WHAT IS THE CORRELATION BETWEEN PRICE AND UNITS SOLD?

OUTPUT:

- **PREMIUM ITEMS: HIGH-PRICED PRODUCTS LIKE MEGA BOARD GAME (\$142.89) SEE A LOWER AVERAGE PURCHASE SIZE OF 1.24 UNITS PER ORDER.**
- **BUDGET ITEMS: LOWER-PRICED PRODUCTS LIKE MEGA MUSIC TOY (\$2.28) AND DINO CASTLE (\$5.00) SCALE UP TO A HIGHER AVERAGE OF 2.74 AND 2.77 UNITS PER ORDER.**

```
67 # CUSTOMER BUYING BEHAVIOUR (PRICE VS UNIT SOLD)
68 • SELECT
69 products.Product_Name,
70 products.Product_Price,
71 ROUND(AVG(sales.Units),2) AS Units_Sold
72 FROM sales
73 INNER JOIN products
74 ON products.Product_ID = sales.Product_ID
75 GROUP BY products.Product_Name,
76 products.Product_Price
77 ORDER BY products.Product_Price DESC;
78
```

Product_Name	Product_Price	Units_Sold
▶ Mega Board Game	142.89	1.24
Rainbow Plush Bear #70	55.51	1.33
Neo Science Lab	51.38	1.44
Pro Science Lab	49.49	1.50
City Board Game	47.95	1.54
Deluxe Card Deck #161	41.86	1.62
Mini Train Set	37.56	1.78
Deluxe Card Deck	36.86	1.72
Rainbow Drone Toy	36.56	1.75
Adventure Robot	35.5	1.77

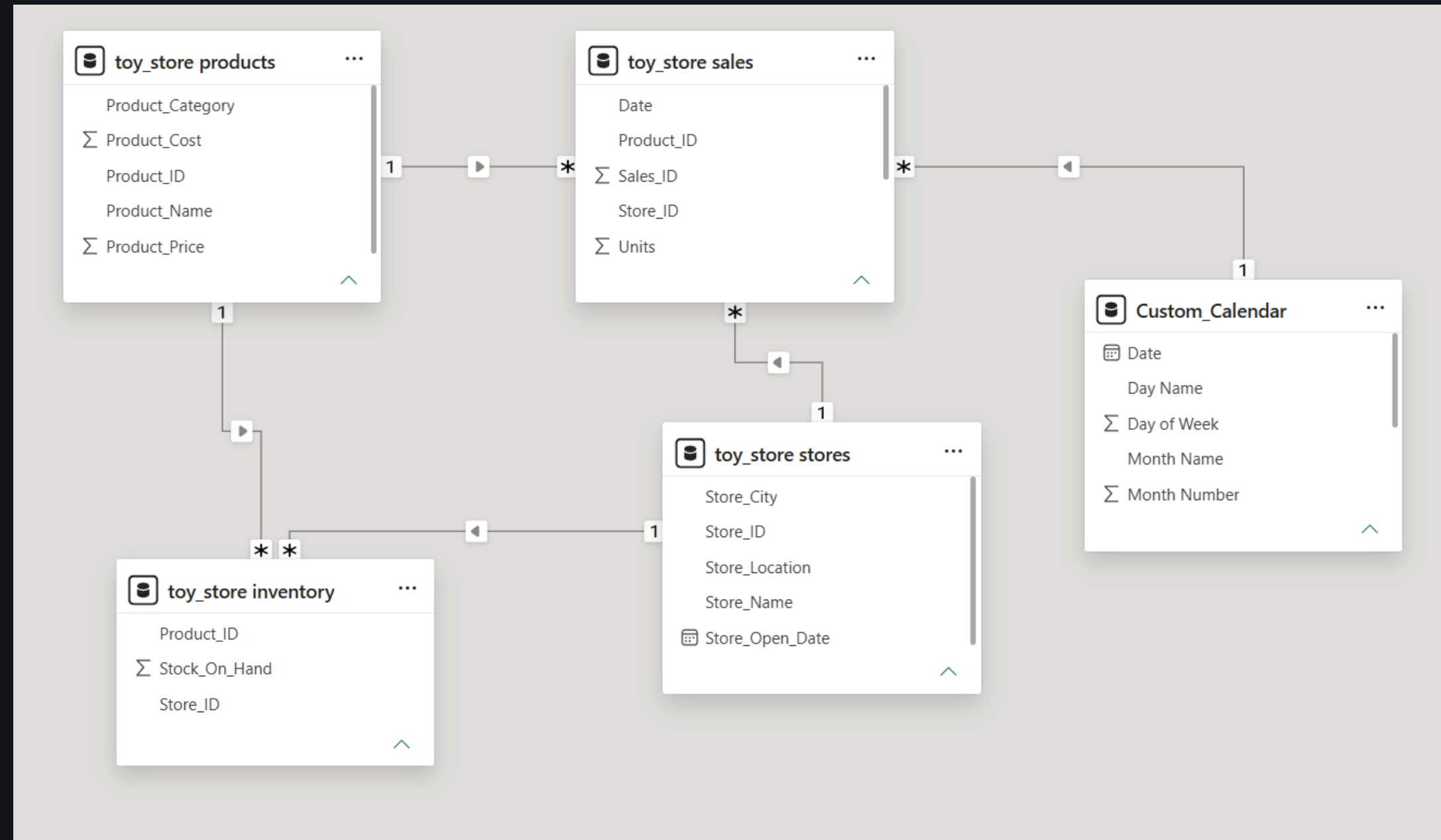
```
67 # CUSTOMER BUYING BEHAVIOUR (PRICE VS UNIT SOLD)
68 • SELECT
69 products.Product_Name,
70 products.Product_Price,
71 ROUND(AVG(sales.Units),2) AS Units_Sold
72 FROM sales
73 INNER JOIN products
74 ON products.Product_ID = sales.Product_ID
75 GROUP BY products.Product_Name,
76 products.Product_Price
77 ORDER BY products.Product_Price DESC;
78
```

Product_Name	Product_Price	Units_Sold
▶ Mega Music Toy	2.28	2.74
Dino Castle	5	2.77
Smart Space Ship	5.03	2.72
Glow Princess Doll	5.33	2.69
Mega Explorer Kit	5.71	2.64
Deluxe Music Toy	5.98	2.68
Smart Music Toy	6	2.72
Mini Robot	6.01	2.67
Eco Art Kit	6.35	2.70
Dino Drone Toy #77	7.24	2.59

DATA MODELING & STAR SCHEMA ARCHITECTURE

DESCRIPTION :

PROCESSED AND CLEANED DATA FROM MYSQL WAS IMPORTED INTO POWER BI. TO OPTIMIZE QUERY PERFORMANCE AND FILTER PROPAGATION, A STAR SCHEMA ARCHITECTURE WAS IMPLEMENTED. THE CENTRAL SALES FACT TABLE WAS LINKED TO THE DIMENSION TABLES (PRODUCTS, STORES, AND INVENTORY) USING ONE-TO-MANY RELATIONSHIPS. ADDITIONALLY, THE CUSTOM CALENDAR TABLE WAS SUCCESSFULLY INTEGRATED INTO THE MODEL BY ESTABLISHING A ACTIVE RELATIONSHIP BETWEEN CALENDAR[DATE] AND SALES[DATE] TO DRIVE ALL TIME-INTELLIGENCE METRICS.

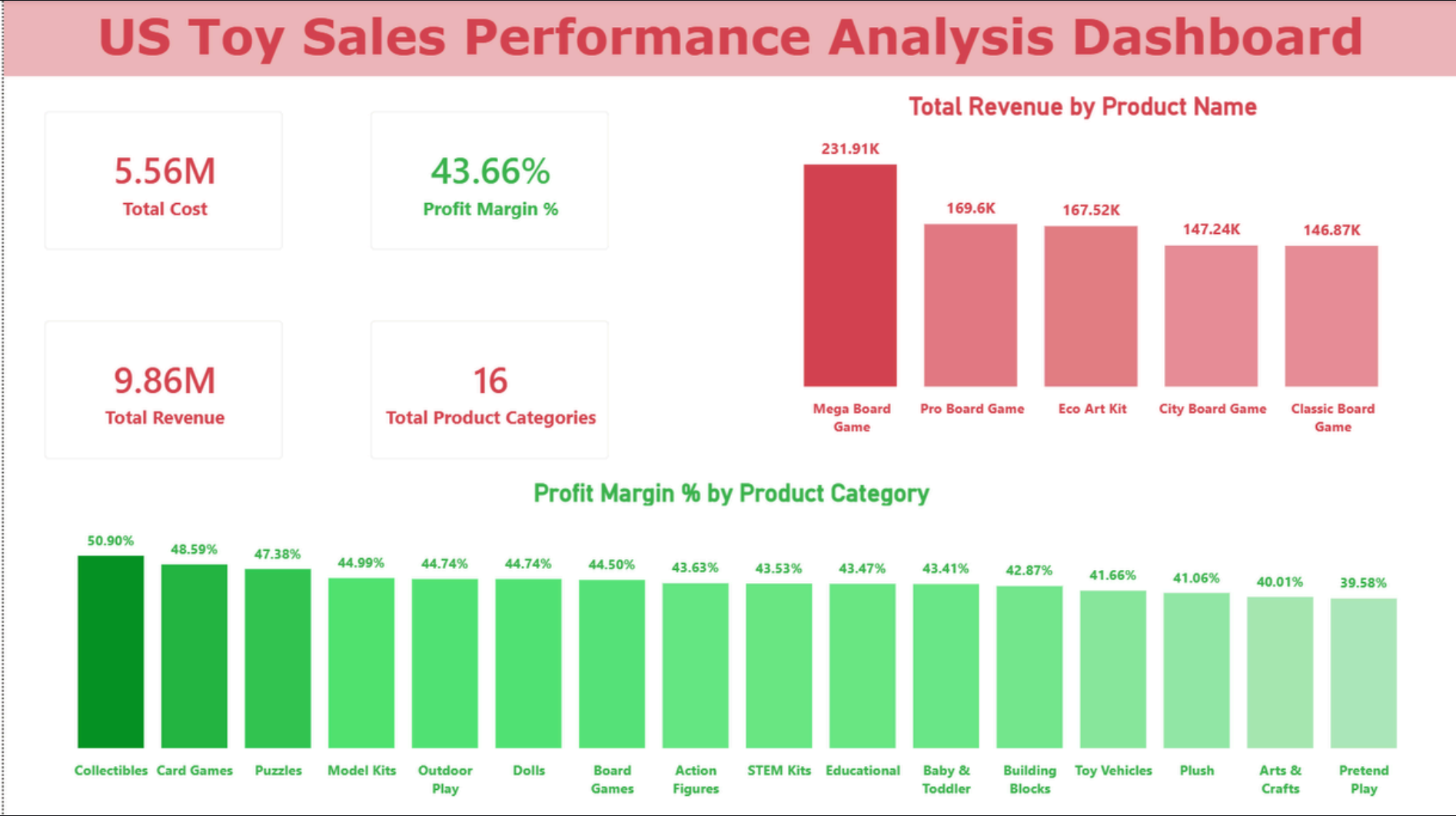


POWER BI DASHBOARD & INSIGHTS

REVENUE & PROFIT MARGIN ANALYSIS

INTERACTIVE INSIGHTS:

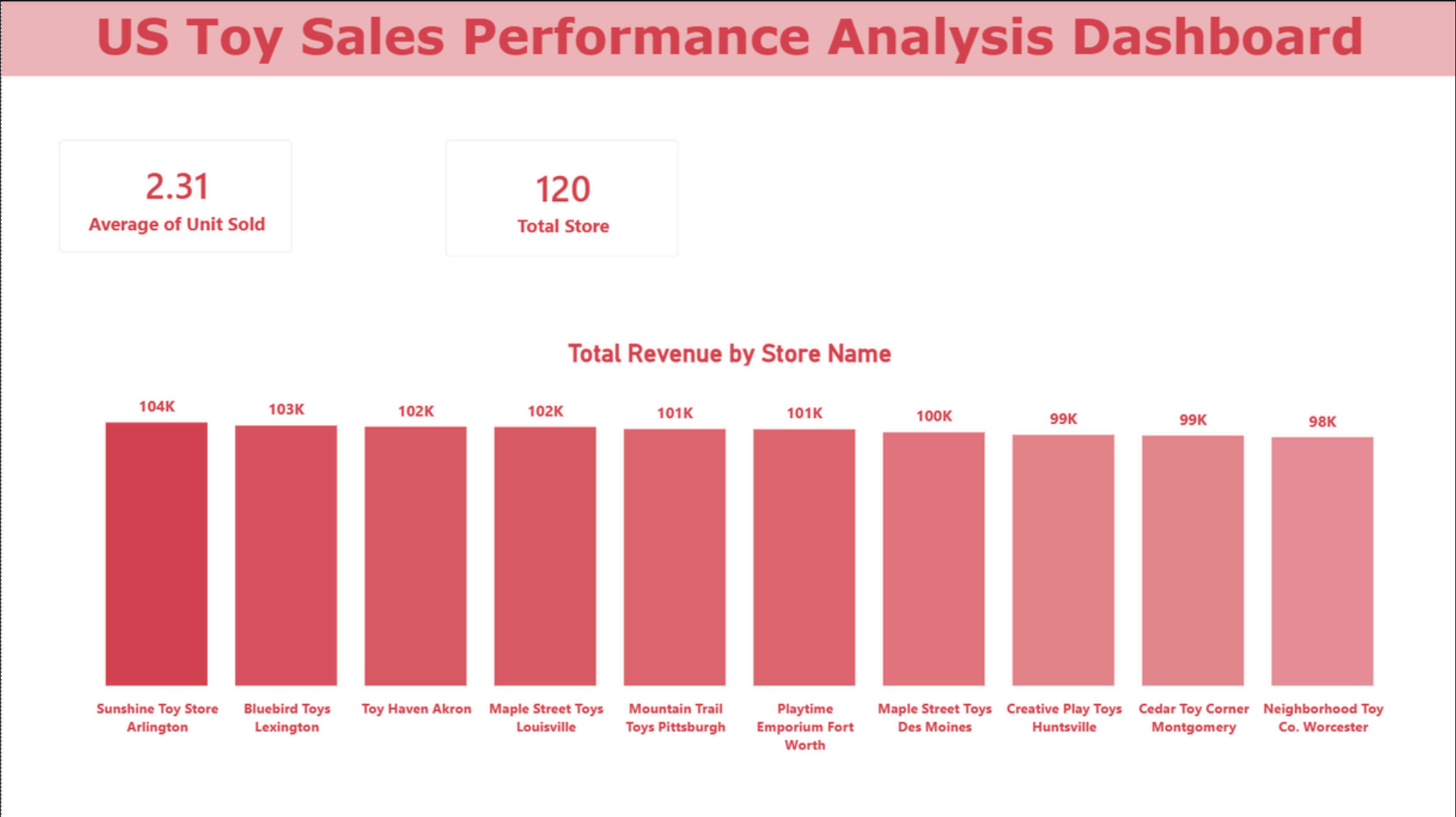
- **TOP PRODUCTS:** MEGA BOARD GAME STANDS OUT AS THE SINGLE HIGHEST REVENUE-GENERATING PRODUCT, PULLING IN APPROXIMATELY \$231.91K.
- **CATEGORY PROFITABILITY:** WHILE MULTIPLE CATEGORIES DRIVE VOLUME, COLLECTIBLES (50.90%) AND CARD GAMES (48.59%) YIELD THE HIGHEST PROFIT MARGINS FOR THE BUSINESS.



POWER BI DASHBOARD & INSIGHTS

STORE REVENUE ANALYSIS

- OUT OF 120 TOTAL OPERATIONAL STORES, SUNSHINE TOY STORE LEADS NATIONWIDE REVENUE, CLOSELY FOLLOWED BY BLUEBIRD TOYS.
- ACROSS ALL RETAIL LOCATIONS, THE AVERAGE BASKET SIZE REMAINS STABLE AT 2.31 UNITS SOLD PER TRANSACTION.

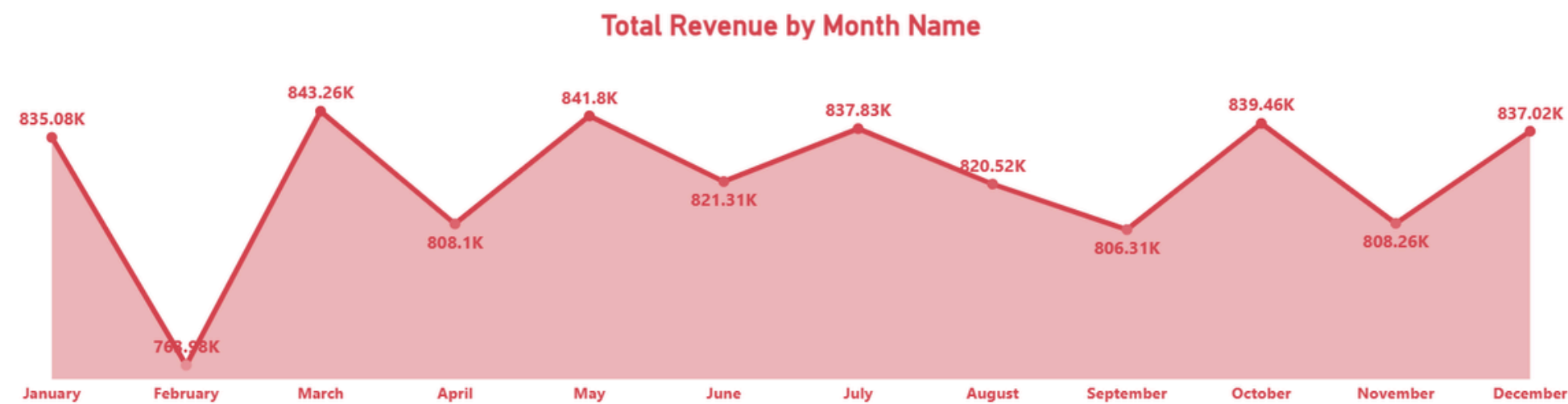


POWER BI DASHBOARD & INSIGHTS

MONTHLY SALES TREND

- **PEAK PERFORMANCE:** SALES PEAK DRAMATICALLY IN MARCH, WITH STRONG, CONSISTENT REVENUE STREAMS ALSO OBSERVED DURING JULY, OCTOBER, AND DECEMBER.
- **LOWEST PERFORMANCE:** FEBRUARY EXPERIENCES THE SHARPEST DIP IN REVENUE ACROSS THE ENTIRE SEASONAL TIMELINE, SHOWING A CLEAR POST-HOLIDAY SLUMP.

US Toy Sales Performance Analysis Dashboard



POWER BI DASHBOARD & INSIGHTS

INVENTORY ANALYSIS (STOCK RISK GRID)

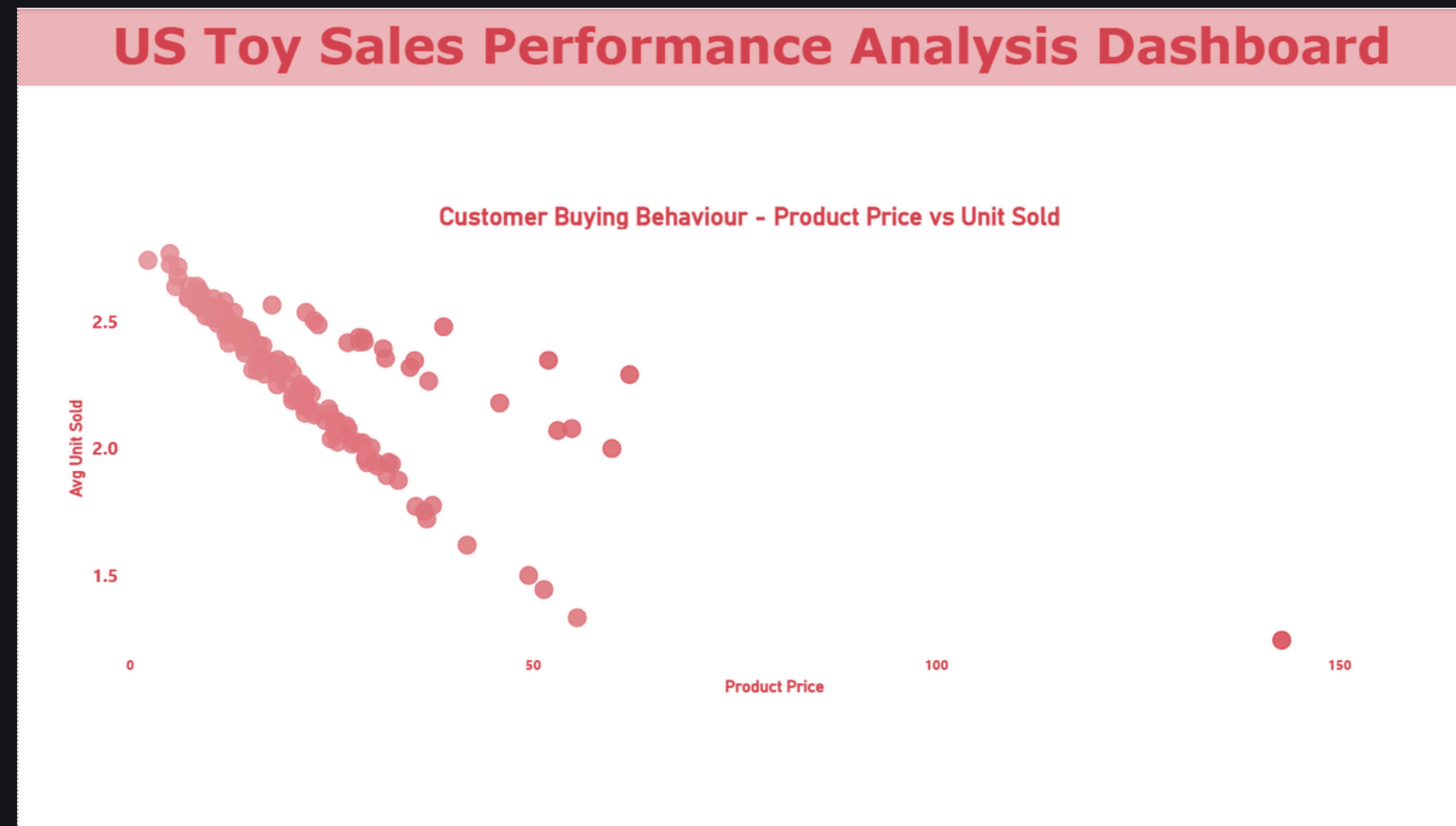
THIS PAGE FEATURES A COMPREHENSIVE MATRIX VISUAL TRACKING INVENTORY RISK. TO ENSURE IMMEDIATE OPERATIONAL UNDERSTANDING, A DIRECT SCREENSHOT IS ATTACHED BELOW TO HIGHLIGHT LOW-STOCK AND HIGH-DEMAND ANOMALIES.

US Toy Sales Performance Analysis Dashboard		
Product_Name	Sum of Stock_On_Hand	Sum of Units
☒ Rainbow Plush Bear #70	375	1744
☒ Neo Science Lab	407	1928
☒ Mega Board Game	421	1623
☒ Pro Science Lab	507	2146
☒ Smart Art Kit	895	2253
☒ Mini Train Set	954	2450
☒ Adventure Robot	1019	2209
☒ Deluxe Card Deck #161	1053	2257
☒ Jungle Action Hero	1077	2507
☒ Rainbow Drone Toy	1094	2543
☒ Deluxe Explorer Kit	1142	2666
☒ City Plush Bear #42	1233	2629
☒ Deluxe Card Deck	1251	2316
☒ Ocean Explorer Kit	1345	2499
☒ Adventure Explorer Kit	1388	2828
☒ Pro Art Kit	1430	2935
☒ Deluxe Science Lab	1438	2515
☒ Galaxy Music Toy	1468	3004
☒ Smart Card Deck	1503	2698
☒ Smart Farm Set #126	1522	2927
☒ Classic Robot #154	1528	3190
☒ Rocket Farm Set	1531	2954
☒ Mini Music Toy	1538	2723
☒ Mega Art Kit #56	1547	2743
☒ Glow Music Toy #35	1565	2459
☒ City Builder Set	1572	2870
☒ Rocket Sticker Book	1580	2696
☒ Mini Explorer Kit	1610	2814
☒ Retro Science Lab	1614	2790

POWER BI DASHBOARD & INSIGHTS

CUSTOMER BUYING BEHAVIOR

- **PRICE SENSITIVITY MATRIX: THE SCATTER PLOT VISUALIZES A CLEAR NEGATIVE DISTRIBUTION TREND IN CONSUMER BUYING HABITS.**
- **INFERENCE: LOW-PRICED (CHEAPER) ITEMS SYSTEMATICALLY DRIVE VOLUME, WITH CUSTOMERS AVERAGING 2 TO 3 UNITS PER PURCHASE. CONVERSELY, PREMIUM (EXPENSIVE) ITEMS ARE STRICTLY BOUGHT AS SINGLE UNITS (1 UNIT PER ORDER).**



CONCLUSION

THIS PROJECT SUCCESSFULLY ANALYZED THE US TOY SALES DATASET BY MOVING FROM RAW DATA TO ACTIONABLE BUSINESS INSIGHTS USING MYSQL AND POWER BI.

- **REVENUE LEADERS: MEGA BOARD GAME PROVED TO BE THE TOP REVENUE GENERATOR, WHILE COLLECTIBLES AND CARD GAMES RECORDED THE HIGHEST PROFIT MARGIN PERCENTAGES FOR THE BUSINESS.**
- **STORE PERFORMANCE: OUT OF 120 STORES, SUNSHINE TOY STORE AND BLUEBIRD TOYS EMERGED AS THE TOP-PERFORMING LOCATIONS.**
- **SEASONALITY: SALES TRENDS PEAKED SHARPLY IN MARCH, WITH ADDITIONAL STRONG REVENUE FLOWS IN JULY, OCTOBER, AND DECEMBER, WHILE FEBRUARY MARKED THE LOWEST SALES PERIOD.**
- **BUYING BEHAVIOR: CLEAR CONSUMER PRICE SENSITIVITY WAS IDENTIFIED, WHERE LOW-PRICED ITEMS ARE CONSISTENTLY BOUGHT IN LARGER QUANTITIES (2 TO 3 UNITS) AND PREMIUM ITEMS ARE STRICTLY PURCHASED AS SINGLE UNITS.**

THANK YOU